NEPAL COLLEGE OF INFORMATION TECHNOLOGY
Assessment_Spring 2026

Level: Bachelor
Programme: BE-CE
Course: **Image Processing and Pattern Recognition**

Year : 2026
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Differentiate digital image processing from computer graphics? Explain briefly about Components of an Image Processing System. 7

   b) What are some basic grey level transformations technique? Perform the contrast stretching to the given image. 8

   52 55 61 66
   63 59 55 60
   62 59 68 65
   63 58 71 69

   Write the stretched image.

2. a) What is histograms equalization? Perform the histogram equalization on the following 8-bit gray level image. 7

3	2	4	5
1	2	2	2
7	3	1	2
7	6	4	5

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OR

Can histogram equalization improve the image poor contrast quality? If yes ,explain with an example.

a) Explain the steps involved in image enhancement in frequency domain. Derive the Kernel for Hadamard Transform of N=4 order. 8

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3. (a) How second order derivative filter is useful in image enhancement? Derive mask for Laplacian second order derivative based filter. 7

(b) Differentiate between image enhancement and image restoration. Explain different noise models in an image with the graphs and equation. 8

4. a) Explain lossless image compression? Briefly explain run length coding with suitable example. 8

b) Perform the Huffman Code for each gray level and find efficiency of the coding in the given image.

.1 2 2 3 0
2 4 4 2 3
4 1 2 2 0
3 4 1 1 2

7

5. a) What is image segmentation and describe region splitting and merging with an example. How does it differ from threshold-based segmentation? 8

a) When do we perform edge detection? What are the edge detection operators with mask. 7

6. a) Explain PCA in the context of feature extraction? Explain how it reduces dimensionality with a suitable diagram or example. 7

b) Define pattern and pattern recognition system. What are the steps use in pattern recognition? Explain 8

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7. Write short notes on(any two) [2*5]
i. Neural network in Image processing
ii. Optimum (Bayes) Statistical Classifiers
iii. Signatures and shape numbers

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National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

Pre-University Examination

Level: Bachelor | Semester : VI_Spring | Year : 2026

Program: B. E Computer | F.M. : 100

Course: **Image Processing and Pattern Recognition** | P. M. : 45

Time : 3 Hrs.

Candidates are required to give their answer in their own words as for as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Image and Digital Image? Explain the Fundamental Steps of Digital Image Processing. 7

   b) Define the uses of histogram? Stretch out the following histogram on gray scale= [0,7] 8

Gray Level	0	1	2	3	4	5	6	7
No. of pixels	20	60	80	90	0	70	40	0

2. a) Find the Original image after being transformed by Hadamard Transform? 7

2	4	6
9	8	7
3	5	1

2 0 1 2

   b) Explain DFT Transform and find the phase angle, power spectrum and Fourier spectrum at point 1,2 of given image? 8

$$F(x)\begin{bmatrix}0\\2\\4\\6\end{bmatrix}$$

"OR"

Derive Haar Transform matrix for kernel=2.

3. a) Explain Image degradation and Restoration Model with its Importances. 8

   b) Differentiate between Lossless and Lossy Compression Techniques with neat diagram too. 7

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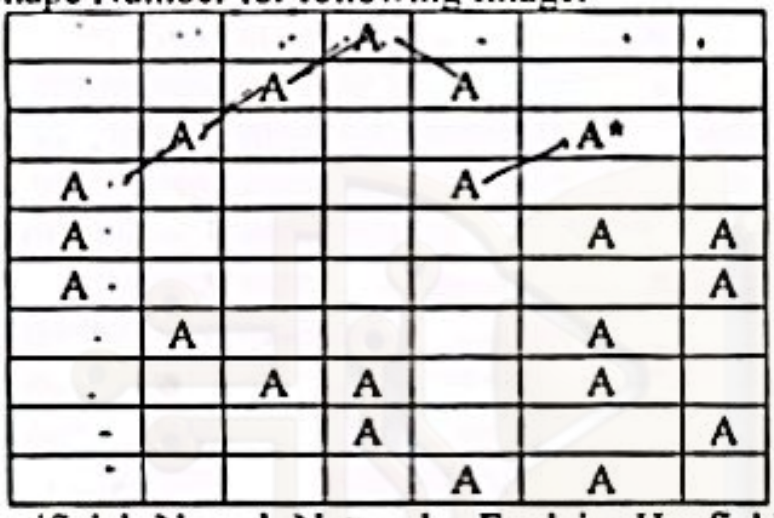
4. a) Explain Image segmentation techniques based on Region Growing, splitting and merging. 8

b) Define Image Segmentation. Explain its types based on Dissimilarities Properties. 7

5. a) Construct Huffman tree for each gray level given and find the efficiency. 8

Gray Level	0	1	2	3	4	5	6	7
No. of pixels	450	1250	650	2034	815	1560	350	1400

b) Find the Shape Number for following Image. 7





6. a) Explain Artificial Neural Network. Explain Hopfield Network in detail. 7

b) Define Pattern and Pattern Class. Explain Steps in Pattern Recognition. 8

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"OR"

Explain SIFT and PCA in detail.

7. Write short notes on following (Any Two) 5x2

a) Sampling and Quantization

b) Run Length Encoding

c) Fast Fourier Transform

Good Luck!!!

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Pokhara University

Everest Engineering College

Final Internal Assessment
Spring-2026

Level: Bachelor — **F.M. 100**

Program: BE Computer — **P.M. 45**

Subject: Image Processing and Pattern Recognition — **Time: 3hrs**

Attempt all the questions.

1 a) Define Digital Image? Explain Fundamental Steps of Digital Image Processing. **7**

b) Explain the image enhancement technique using magnification with replication and interpolation. **8**

2 a) What is histogram equalization? Perform the histogram equalization on the following 8-bit gray level image. **7**

3	2	4	5
1	2	2	2
7	3	1	2
7	6	4	5

OR

Can histogram equalization improve the image poor contrast quality? If yes, explain with an example.

b) Explain DFT Transform and find the phase angle, power spectrum and Fourier spectrum at point 0, 1 of given image? **8**

$$\bar{F}(x)\begin{bmatrix}0\\2\\4\\6\end{bmatrix}$$

3 a) What are the properties of Haar Transform? Find Haar matrix for N=4. **8**

b) What is Noise in image? Explain different type of noise model with necessary equation and curve. **7**

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4 a) Define Data and Coding redundancy. Suppose a source generates the symbols s1, s2, s3, s4, s5 randomly with probability p1=0.4, p2= 0.2, p3= 0.2, p4= 0.1 and p5 = 0.1 respectively. Generate the codeword for each symbol using Huffman coding and also calculate entropy and efficiency. 8

b) Explain why image compression is necessary? Explain the working principle of lossy predictive coding with necessary block diagram. 7

5 a) Define chain code, signature and shape number. Explain 4 and 8 connectivity segments to explain chain code with suitable examples. 7

b) Briefly explain different methods of selecting threshold value for image segmentation? 8

6 a) Define Image Segmentation. Perform Region Splitting for the following given image Threshold (T) ≤ 3. 8

5	6	6	6	7	7	6	6
6	7	6	7	5	5	4	7
6	6	4	4	3	2	5	6
5	4	5	4	2	3	4	6
0	3	2	3	3	2	4	7
0	0	0	0	2	2	5	6
1	1	0	1	0	3	4	4
1	0	1	0	2	3	5	4

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b) What do you mean by scale invariant feature transform. Explain the step for SIFT algorithm. 7

7 Write short notes on: (Any two) 2*5 =10

a) PCA
b) Hough Transform
c) Image Sampling and Quantization

Good Luck

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POKHARA ENGINEERING COLLEGE
Internal Assessment Examination

Level: Bachelor | Semester – Spring | Year . 2026
Programme: BCE | Full Marks: 100
Course: Image Processing and Pattern Recognition | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define fundamental steps in DIP with block diagram 8

   b) Equalize the histogram from the given data. 7

r_k	0	1	2	3	4	5	6	7
n_k	5320	1000	500	526	1236	956	856	128

2. a) Define point operation. Explain the technique used by Bit plane intensity level slicing for the purpose of image enhancement. 8

   b) What is zooming? Illustrate with an example by interpolation and replication. 7

3. a) What do you mean by image enhancement in frequency domain? What are the properties of 2D Fourier Transform? 8

   b) Compute the Haar transformation for order N = 2.

   Or

   Derive the equations of Fast Fourier Transformation 7

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4. a) Explain image degradation model. 8

   b) What is periodic noise? Explain any two band reject filters to remove periodic noise in frequency domain. 7

5. a) Explain region splitting and region growing in image segmentation with an algorithm. 8

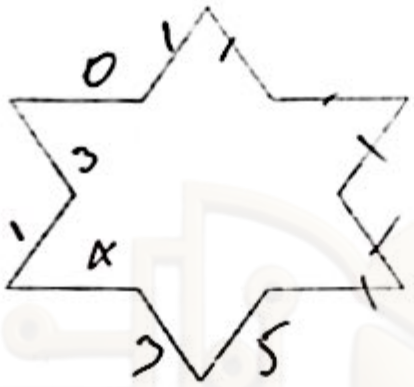
   b) Explain Huffman coding. Determine Huffman code for given image data and calculate the efficiency. 7

Gray levels	0	1	2	3	4	5	6	7
Frequency	7	13	23	44	58	28	16	11

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6. a) Define chain codes. Find the shape number for the given figure. 8



7

b) Briefly explain the various steps of pattern recognition.

7. Write short notes on any two: 2×5

a) Fourier Descriptors

b) Thresholding

c) Prewitt and Sobel operator

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GANDAKI COLLEGE OF EINGINEERING AND SCIENCE

Level: Bachelor | Semester: Spring | Year : 2026
Programme: BE CE VI | Full Marks: 100
Course: **Image Processing and Pattern Recognition** | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is a digital image? Describe the basic stages involved in a Digital Image Processing (DIP) system with a neat block diagram. 7

   b) Describe spatial-domain low-pass (smoothing) filters used in image enhancement. Explain their working principle with suitable examples. 8

2. a) Equalize the given Histogram. 7

Gray Level	0	1	2	3	4	5	6	7
No.of Pixel	790	1023	850	656	329	245	122	81

b) Explain Frequency domain filtering Operation in digital image processing. List out any four properties of Fourier Transform. 8

3. a) Explain Hadamard Transform in Image Processing. List out the common properties of Haar Transform. 7

   b) What is image noise? Explain the different noise models used in digital image processing with suitable mathematical equations and probability distribution curves. 8

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4. a) What is image compression? Differentiate between lossless and lossy image compression techniques with their advantages, disadvantages, and applications. 7

b) Explain LZW Coding with Example. 8

OR

Consider a source with 7 Message Having Probabilities
0.25,0.25,0.125,0.125,0.125,0.0625,0.0625
1.Find Average Code Length
2. Entropy
3. Efficiency
4. Redundancy

5. a) Define Image Analysis. Explain Local and Global thresholding with example. 7

b) What is chain code? How to you normalized chain code? 8

OR

What is a gradient filter? Explain the Sobel gradient filter in detail along with its algorithm for implementation.

6. a) Explain how a Neural Network is used for image recognition. Illustrate the working of a simple perceptron with a neat diagram. 7

b) Describe the architecture of a Pattern Recognition System with the help of a block diagram and explain the function of each component. 8

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7. Write short notes on **(Any Two)** 2×5

a) Scale-Invariant Feature Transform (SIFT).
b) Quad Tree representation
c) Sampling and Quantization.

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POKHARA UNIVERSITY
FACULTY OF SCIENCE AND TECHNOLOGY
SCHOOL OF ENGINEERING

Exam	Final Internal Examination 2026 Spring		
Level	B.E.	F.M	100
Program	Computer	P.M	45
Year/ Part	III/II	Time	3 Hrs

Subject: Image Processing and Pattern Recognition

Candidates are required to give answers in their own words as far as practicable.
The figure in the margin indicates full marks. Assume suitable data if necessary.
Attempt all the questions.

1 a) Explain the fundamental steps of Digital Image Processing with a neat block diagram. 7

b) Differentiate between sampling and quantization. Explain the 4, 8 and m connectivity of pixels with the help of an example. 8

2 a) Explain histogram equalization. How does it improve image contrast? Compute the histogram equalization of the given image. 8

Gray levels	0	1	2	3	4	5	6	7
No of Pixels	5320	1000	500	525	1236	956	856	128

b) Differentiate between spatial-domain filtering and frequency-domain filtering. Explain any two frequency-domain smoothing filters. 7

3 a) Describe Hadamard transform. Compute the Hadamard transformation for the kernel of order N=2. 7

b) Explain Salt and pepper noise? Which filter is used to remove this noise? Explain Adaptive mean filter. 8

4 a) Consider a string 'wabbawabbawoo' which we would like to encode using the LZW algorithm. Construct the dictionary D and Determine the encoded string using the minimum number of bits needed to represent all codewords. 7

b) Draw a huffman tree. Also Calculate the compression ratio, entropy and efficiency from the image below. 8

Character	a	b	c	d	e	f
Frequency	5	9	12	13	16	45

5 a) Explain region growing segmentation. Explain seeded regions growing with an example. Discuss the advantages and limitations of the method. 7

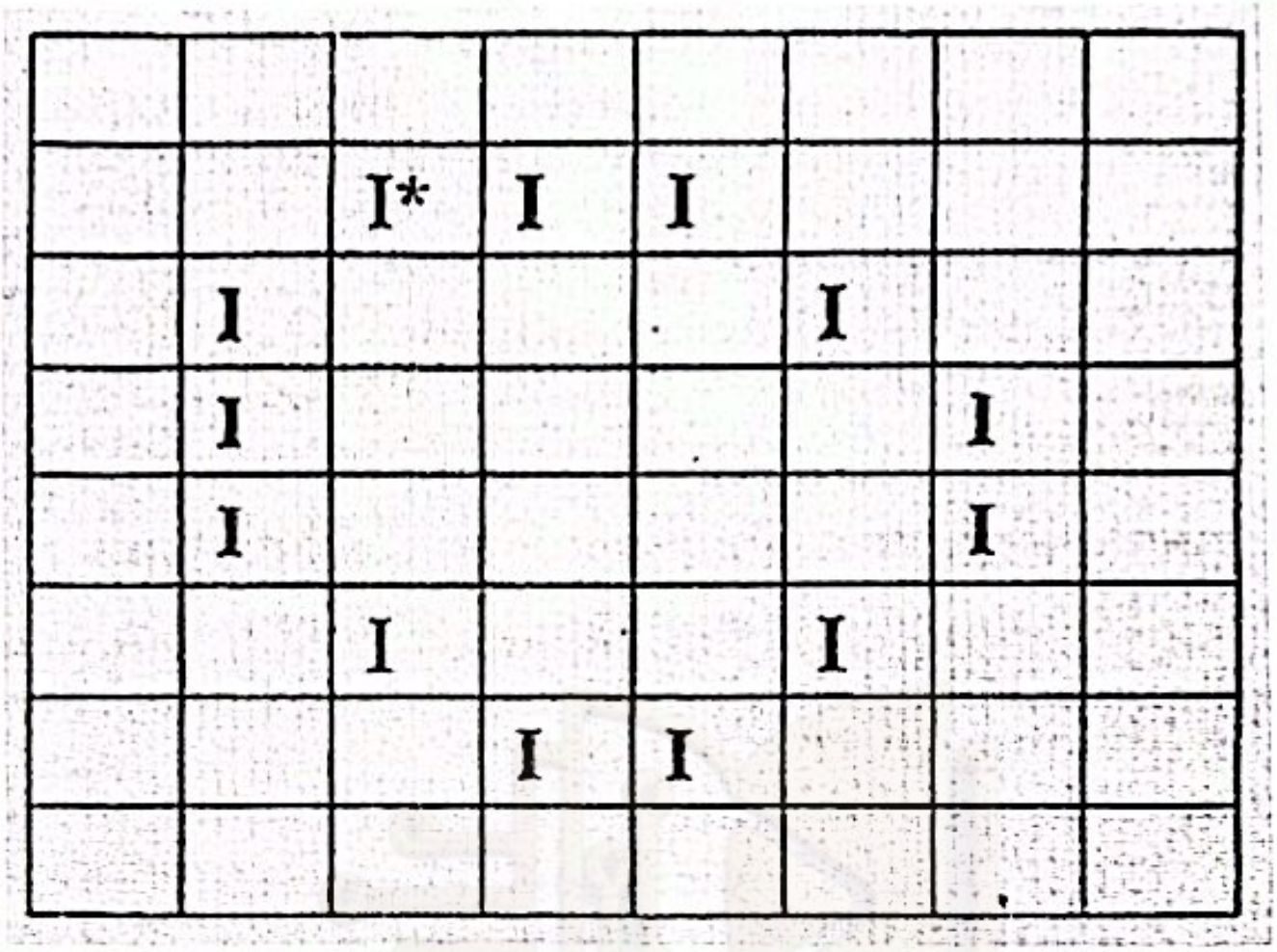
b) Explain the need for image compression. Differentiate between lossy and lossless compression techniques with examples. 8

-- Best of luck --

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6 a) Describe polygon approximation techniques used for boundary representation. Given an image, I* represents the starting point. Write down the 8-chain code and find the shape number. 7



b) What is PCA? Explain the steps involved in Principal Component Analysis (PCA) for feature extraction. Also how it reduces dimensionality with a suitable diagram or example.

OR

What is the Scale-Invariant Feature Transform (SIFT)? Explain how it helps in detecting and describing key points. Why is it considered invariant to scale and rotation? 8

7 Write Short notes. (Any two) 2X5

a) LZW coding

b) Point Detection

c) Perceptron

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GANDAKI COLLEGE OF EINGINEERING AND SCIENCE

Level: Bachelor | Semester: Spring | Year : 2026
Programme: BE CE VI | Full Marks: 100
Course: **Image Processing and Pattern Recognition** | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) What is a digital image? Describe the basic stages involved in a Digital Image Processing (DIP) system with a neat block diagram. 7

   b) Describe spatial-domain low-pass (smoothing) filters used in image enhancement. Explain their working principle with suitable examples. 8

2. a) Equalize the given Histogram. 7

Gray Level	0	1	2	3	4	5	6	7
No.of Pixel	790	1023	850	656	329	245	122	81

   b) Explain Frequency domain filtering Operation in digital image processing. List out any four properties of Fourier Transform. 8

3. a) Explain Hadamard Transform in Image Processing. List out the common properties of Haar Transform. 7

   b) What is image noise? Explain the different noise models used in digital image processing with suitable mathematical equations and probability distribution curves. 8

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4. a) What is image compression? Differentiate between lossless and lossy image compression techniques with their advantages, disadvantages, and applications. 7

b) Explain LZW Coding with Example. 8

OR

Consider a source with 7 Message Having Probabilities
0.25,0.25,0.125,0.125,0.125,0.0625,0.0625
1.Find Average Code Length
2. Entropy
3. Efficiency
4. Redundancy

5. a) Define Image Analysis. Explain Local and Global thresholding with example. 7

b) What is chain code? How to you normalized chain code? 8

OR

What is a gradient filter? Explain the Sobel gradient filter in detail along with its algorithm for implementation.

6. a) Explain how a Neural Network is used for image recognition. Illustrate the working of a simple perceptron with a neat diagram. 7

b) Describe the architecture of a Pattern Recognition System with the help of a block diagram and explain the function of each component. 8

7. Write short notes on **(Any Two)** 2×5

a) Scale-Invariant Feature Transform (SIFT).
b) Quad Tree representation
c) Sampling and Quantization.

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UNITED TECHNICAL COLLEGE

Semester-Spring

Subject: Image Processing and Pattern Recognition — Full Marks: 100

Time: 3 hrs — Pass Marks: 45

Attempt all Questions:

1. a) Define point operation. Explain the process of digital negative and bit plane slicing for the purpose of image enhancement. [8]

b) Define digital image. Explain the process of image sampling and quantization in the digital image representation with figures. [7].

2. a) What is histogram equalization? Perform the histogram equalization on the following 8-bit gray level image. [8]

3	2	4	5
1	2	2	2
7	3	1	2
7	6	4	5

OR

. List the basic gray level transformation technique. Perform the contrast stretching on the 3-bit gray level image given below.

5	6	2	3
6	5	4	7
2	2	4	1
2	3	6	4

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b) What is image restoration? Explain the model of the image degradation and restoration process with a block diagram. [7]

3. a) Write the algorithm of Haar transform. Compute the Haar Transformation for the order N=2. [8]

b) Compute the Hadamard transform of the image given below. [7]

1	1	2	1
2	1	1	2
1	3	2	1
2	1	2	1

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4. a) What is coding redundancy? Construct Huffman coding for each gray level and find the compression ratio and coding efficiency. [8]

Gray level (rk)	0	1	2	3	4	5	6	7
Frequency	500	200	250	100	250	500	150	50

b) Define image compression. Explain lossless predictive coding with necessary diagram. [7]

5. a) Define thresholding. How can we select the efficient threshold value for image segmentation? Mention limitations of single valued thresholding for global thresholding. How adaptive thresholding solve this issue? Explain. [8]

OR

What is region-based segmentation? Explain the segmentation method based on region splitting and merging technique with an example.

b) What is pattern recognition? Explain the various steps of pattern recognition system [7]

6. a) Define pattern and pattern class. Explain Baye's classifier for pattern recognition. [8]

b) Given an image, "A" represents its pixel position. If A* is the starting pixel, write down the 8-chain code and find shape number of it. [7]

			A*	A		
		A			A	
		A		A		
	A				A	
		A	A	A		

7. Write short notes on: (Any Two) [2*5=10]

a) Hopfield Network

b) Edge detection

c) Median filter

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Universal Engineering & Science College

Affiliated to Pokhara University

Chakupat, Lalitpur

Level: Bachelor | Semester: 6th | Year : 2026
Programme : BE Computer | Time : 3 hours | Full Marks: 100
Subject: Image Processing & Pattern Recognition | Pass Marks: 45

Pre-Board Examination- 2083(Spring 2026)

Candidates are required to give their answers in their own words as practicable. The figures in the margin indicate full marks.

Attempt All the questions:

1. a) Define Sampling and Quantization in image acquisition. How adjacency of the pixels affects connectivity and path between the pixels? Explain with suitable Examples. [8]

b) What is Histograms Equalization? Perform the histogram equalization on the following 8-bit gray level image and also scale the intensity from 1 to 20. [7]

3	2	4	5
1	2	1	2
7	3	1	2
7	6	4	7

2. a)What are the steps involved in frequency domain filtering? Explain the essential properties of 2D Discrete Fourier Transform. [8]

b)Why do we use Haar Transform in image processing? Explain its algorithm for the order N=2. [7]

OR,

Compute 2D discrete Walsh-Hadamard Transform for the given image block below:

5	6	8	10
6	6	5	7
4	5	3	6
7	8	3	5

3. a) Explain different noise models in an image with the graphs and their PDF. [8]

b) Explain Image Restoration and Degradation model with necessary diagrams. [7]

4. a) Define Image Compression. Explain Lossless and Lossy Image Compression Techniques. [7]

b)What is Interpixel Redundancy? Compress the string "SHAISAMSHAIHAI" using LZW coding. [8]

OR,

From the following probability distribution construct the Huffman Code for each gray level and find efficiency of the coding.

r_k	0	1	2	3	4	5
$P(r_k)$	0.05	0.1	0.25	0.15	0.3	0.15

5. a) Why image segmentation is necessary in image processing? Explain segmentation by region splitting and merging algorithm. [8]

OR,

How Edge linking is used in boundary detection of an image? Explain Hough Transform with suitable figures.

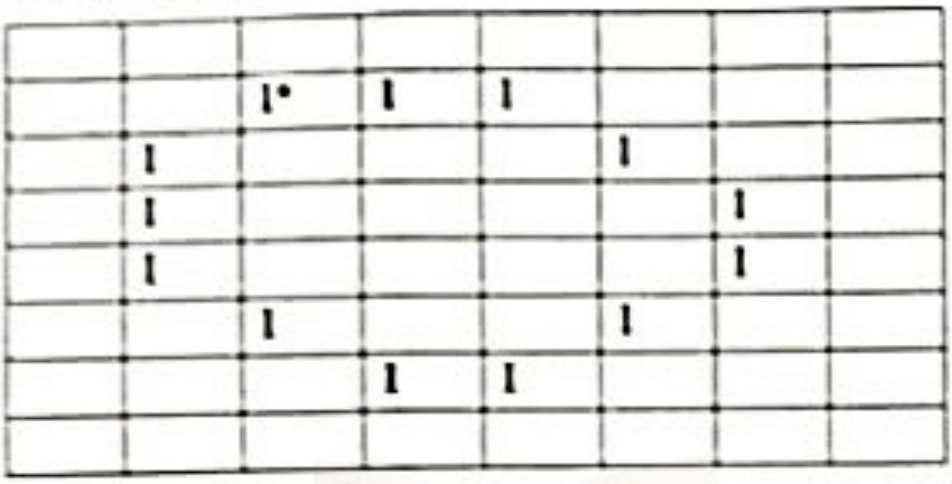
b) What is Local and Global Thresholding? Explain the algorithm for Optimum Thresholding using Otsu's method. [7]

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6. a) Explain Crack code and Signature. Also find the Chain code and Shape number for the following image. 1* is the starting point and the rotation is clockwise direction. [7]



b) What is pattern? Explain Pattern classes in brief. Also describe the steps of pattern recognition system with its block diagram. [8]

OR

Explain Scale-Invariant Feature Transform (SIFT) in details also write how SIFT keypoints can be used for Feature extraction and classification

7. Write short notes on: **(Any Two)** [5*2]

a) Discrete Cosine Transform

b) Fourier Descriptor

c) Bayes Statistical Classifier

d) Perceptron

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National Academy of Science and Technology

(Affiliated to Pokhara University)

Dhangadhi, Kailali

First Terminal Examination

Level: Bachelor | Semester : VI_Spring | Year : 2026

Program: B.E Computer | F.M. : 100

Course: **Image Processing and Pattern Recognition** | P. M. : 45

Time : 3 Hrs.

Candidates are required to give their answer in their own words as for as practicable. The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Define Digital Image Processing? Explain the Fundamental Steps in Digital Image Processing. 7

   b) What are applications of histogram? Equalize the following histogram on gray scale= [0,7] 8

1	3	3	2
4	2	1	3
3	1	1	5
3	4	2	5

2. a) List and explain any two Different Gray Level transformation. 7

   b) Explain DFT Transform and find the phase angle, power spectrum and Fourier spectrum at point 0,1 of given image? 8

$$F(x) \begin{bmatrix} 0 \\ 2 \\ 4 \\ 6 \end{bmatrix}$$

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"OR"

Find DCT for F(x)=

52	52	52	52
55	55	55	55
61	61	61	61
66	66	66	66

3. a) Explain Image degradation and Restoration Model with its Importances. 8

   b) Explain Sampling and Quantization with its Importance in Image Processing. 7

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4. a) Explain Hadamard Kernel by Deriving the Kernel for N=4. 8

"OR"

Find the Original image by performing Hadamard Transform on Given Image.

F(x)=

5	15	4	3
8	11	10	2
6	7	8	9
91	81	71	61

b) Gray level histogram of an image is shown in 3-bit system is as Follows. Expand the Stretching of histogram over the entire range. 7

Gary level	0	1	2	3	4	5	6	7
Frequency	0	0	50	60	50	20	10	0

5. a) Explain the Concept of Zooming in Image Enhancement in Spatial Domain. 8

b) Find the original image after being transformed using Haar Transformation.

F(x)=

5	10	15	20
8	1	1	2
77	7	0	1
0	8	0	7

7

6. a) Explain Smoothening and Sharpening Spatial Filters in Image Processing. 7

b) Define Image and Digital Image? Explain the elements of DIP. 8

7. Write short notes on following (Any Two) 5x2

a) Salt and Pepper Noise
b) Elements of Visual Perception
c) Fast Fourier Transform

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Good Luck!!!

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test I

Date: 2083/03/1			
Level	BE	Full Marks	50
Programme	BCE	Time	
Semester	VI	1.5 hrs	

Subject: - Image Processing and Pattern Recognition

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt All questions. Each question carries equal marks.*
- ✓ *The figures in the margin indicate Full Marks.*
- ✓ *Assume suitable data if necessary.*

1 a) Define Digital Image Processing? Explain the elements of Digital Image Processing Systems. 8

b) Given the following gray level histogram of an image. Compute the gray levels after histogram equalization. 7

M	0	1	2	3	4	5	6	7
Gray Level	100	300	400	500	600	20	180	400

2 a) What is neighborhood operations in image enhancement? Derive Laplacian Filter Mask by using second order partial differential equation. 8

b) What is Zooming? Illustrate with an example by interpolation and replication. 7

3 a) Why do we use Haar Transform in Image Processing? Explain its algorithm for the order N = 2. 8

b) What is interpixel redundancy? Compress the string "SHAISHAMSHAIHAI" using LZW coding. 7

4 Write short notes (any one) 5

a) Bit plane slicing

b) Lossy and Lossless compression

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COSMOS COLLEGE
OF MANAGEMENT AND TECHNOLOGY

Term Test II

Date: 2083/04/10			
Level	BE	Full Marks	50
Programme	BEIT	Time	
Semester	VI	1.5 hrs	

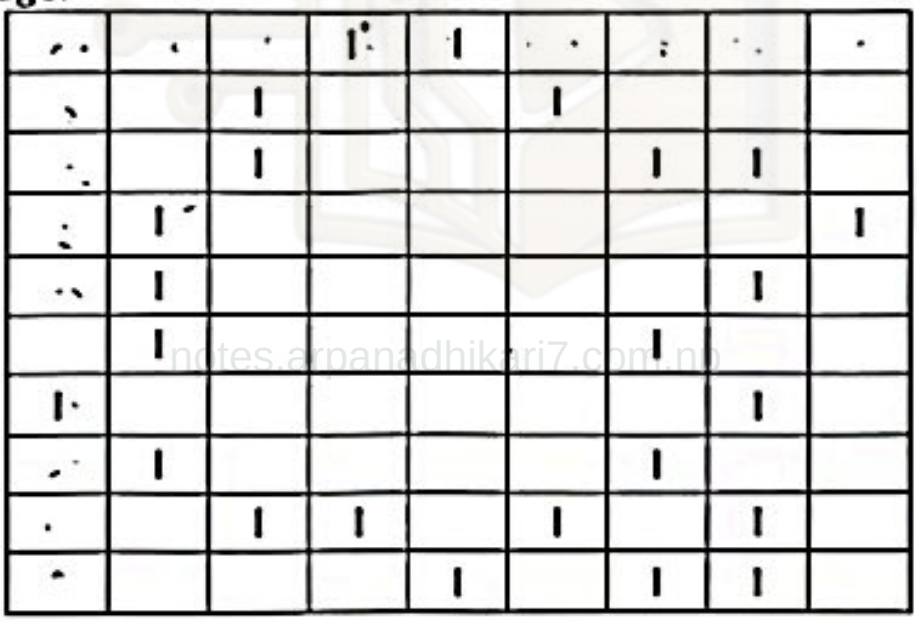
Subject: - Image Processing and Pattern Recognition

- ✓ *Candidates are required to give their answers in their own words as far as practicable.*
- ✓ *Attempt All questions.*
- ✓ *The figures in the margin indicate Full Marks.*
- ✓ *Assume suitable data if necessary.*

1 a) Why Image segmentation is necessary in image processing? Explain segmentation by region splitting and merging algorithms. 8

b) Explain Image Restoration and Degradation model with necessary diagrams. 7

2 a) Define chain code and shape number. Also fine the shape number for the following Boundary Image. 8



b) Define PCA. Explain how features of an image are extracted using PCA. 7

3 a) Define thresholding. How can we select the efficient threshold value for image Segmentation? Mention limitations of single valued 8

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Madan Bhandari College of Engineering

Urlabari-3, Morang

Final Internal Examination (Spring 2026)

Level: Bachelor | Full Marks: 60

Programme: B.E. Computer | Pass Marks: 30

Year/Part: III/II | Time: 2 hrs

Subject: - Image Processing and Pattern Recognition

- ✓ *Candidates are required to give their answers in their own words as far as possible.*
- ✓ *Attempt all the Questions*

1. **a)** Define neighboring of pixels. Explain its types with example of each type.

   b) Find Huffman codes from given histogram information of an image.

Grey Level	0	1	2	3	4	5	6	7
No. of Pixels	35	30	15	10	38	80	38	0

2. **a)** Consider the following image with the new pixel at (2,2) if the smoothing is done using a 3*3 bit neighborhood. Find
   i. Mean Filter
   ii. Weighted average filter
   iii. Median Filter
   iv. Min Filter
   v. Max Filter

1	2	1	3	8
3	8	8	5	1
5	7	5	4	4
2	2	1	8	6
6	5	3	3	6

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   b) Write algorithm of Haar Transform. Compute the Haar Transformation for the order N=2.

3. **a)** Define salt-and-pepper noise. Which spatial filter is used to reduce salt-and-pepper noise? Why?
   b) Explain, how you can use Hamming neural network for object recognition. Explain the process of single layer perceptron.

4. Write short notes: (Any three)

a) Steps in Pattern Recognition

b) Lossy Image Compression

c) Neural Network

d) Zooming

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